Tanya Parekh

Mechatronics Engineer

RESUME_1229

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Total Experience: 8 years

PROFESSIONAL SUMMARY

Mechatronics Engineer with 8 years of experience in product development, using CAD, SolidWorks, Arduino, ESP32. Experienced in electromechanical systems, integration tests and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Control Systems, Sensors, C++, Python, Arduino, PLC, Testing, MATLAB, ESP32, SolidWorks, Motor Drivers, CAD

Methods: Electromechanical systems, Integration tests, Product prototyping

PROFESSIONAL EXPERIENCE

Senior Mechatronics Engineer | JuniperWorks Systems | Jan 2023 - Present

- Improved prototype throughput by 39% through actuator timing optimization
- Validated 485 integration test cases across mechanical and electrical interfaces
- Reduced assembly cost by 21% after redesigning electromechanical fixtures
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Mechatronics Engineer | BrioTech Labs | Jan 2021 - Dec 2022

- Reduced assembly cost by 40% after redesigning electromechanical fixtures
- Raised measured actuator positioning consistency by 30% after calibration

Associate Mechatronics Engineer | HelioNex Labs | Jul 2018 - Dec 2020

- Improved prototype throughput by 31% through actuator timing optimization
- Validated 279 integration test cases across mechanical and electrical interfaces

PROJECTS

Smart Dispensing Unit | Testing, Sensors, C++, Control Systems

- Built controller firmware, mechanical enclosure and fault signals.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

Lab Instrument Positioner | Testing, Sensors, Control Systems, C++

- Designed stepper motion stages and calibration routines.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

EDUCATION

B.Tech in Mechatronics Engineering, Bayview Institute of Engineering, 2018
Relevant Methods: Electromechanical systems, Integration tests, Product prototyping

CERTIFICATIONS

Mechatronics Systems Integration (2022)